



Republic of Rwanda
Ministry of Education



LSVRS502

REMOTE SENSING TECHNOLOGY

Apply Remote Sensing Technology

Competence

RQF Level:

Level 5

Learning Hours



80

Credits:

8

Sector:

Construction and Building Services

Trade:

Land Surveying

Module Type:

Specific

Curriculum:

CBSLSV5002-TVET Certificate V in Land Surveying

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Purpose statement	This module describes the knowledge, skills, and attitudes required to apply remote sensing technology. At the end of this module, the learner will be able to plan the work, collect remote sensing data, and produce plans and models.					
Learning assumed to be in place	Land law and regulations, surveying lab equipment maintenance, Applied Physics, Applied mathematics					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	Group project and presentation	20%				
	Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competence and Performance Criteria

Elements of competence	Performance criteria
Plan the work	1.1. Remote sensing systems are properly identified based on the work requirement
	1.2. Site Reconnaissance is properly conducted based on the environmental conditions
	1.3. Legal and regulatory requirements are properly described according to their types
	1.4. Resources are properly allocated according to planned activities.
Collect remote sensing data	2.1. Data collection methods are correctly selected according to the work requirements
	2.2. Control points are accurately established based on the scope of the work
	2.3. Data are accurately captured based on the selected method
Produce maps and models	3.1. Data are accurately pre-processed according to the procedures
	3.2. Data are accurately processed according to the types of products to be generated
	3.3. Maps and models are accurately generated according to their formats

Intended Knowledge, Skills, and Attitude

Knowledge	Skills	Attitude
✓ Describing work planning procedure ✓ Describing legal framework governing Photogrammetry and remote sensing application ✓ Describing Ethical Considerations ✓ Describing surveying mathematics ✓ Identifying remote sensing and photogrammetric surveying tools and equipment ✓ Land rules and regulations.	✓ Designing the work plan ✓ Applying environmental management skills ✓ Operating photogrammetric surveying tools and equipment ✓ Operating Remote sensing tools and equipment ✓ Applying GIS skills ✓ Applying Image processing techniques ✓ Applying communication skills ✓ Using Remote Sensing Software ✓ Using Photogrammetry Software	✓ Having critical thinking ✓ Having Team working spirit ✓ Having responsible ✓ Being Persistent ✓ Having Time management spirit ✓ Being Polite ✓ Being Decisive ✓ Being Research oriented ✓ Being Innovative ✓ Being Flexible ✓ Being Attentive ✓ Being Detail-oriented ✓ Being Prompt ✓ Being Confident ✓ Having Problem solving attitude

	✓ Performing Photogrammetric Data Collection ✓ Performing Remote sensing data Collection ✓ Applying ICT Skills	
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Learning outcomes	At the end of the module the learner will be able to: 1. Plan the work 2. Collect Remote Sensing data 3. Produce maps and models
Learning outcome 1: Plan the work	Learning hours: 20 hours

Indicative content

- **Identifying remote sensing systems**
 - ✓ **Elements of remote sensing**
 - ✚ Identify Energy source
 - ✚ Describe EMR interaction with atmosphere
 - ✚ Analyse EMR Interaction with targets
 - ✚ Recording of energy by Sensor
 - ✚ Transmit information from the sensors
 - ✚ Interpret information
 - ✓ **Types of Remote sensing sensors**
 - ✚ Optical Sensors
 - ✚ Radar Sensors
 - ✚ Lidar Sensors
 - ✚ Multispectral Sensors
 - ✚ Hyperspectral Sensors
 - ✓ **Types of remote sensing platforms**
 - ✚ Space borne platforms (Low Earth Orbit (LEO) Satellites, Medium Earth Orbit (MEO) Satellites and Geostationary Orbit (GEO) Satellites)
 - ✚ Airborne Platforms
 - ✚ Ground-Based Platforms
 - ✓ **Types of Photogrammetric surveying**
 - ✚ Aerial Photogrammetry
 - ✚ Terrestrial Photogrammetry
 - ✚ Close range Photogrammetry

- **Conducting site reconnaissance**
 - ✓ Prepare site
 - ✓ Conduct site visit
 - ✓ Elaborate site reconnaissance report
- **Identifying Legal and regulatory requirements**
 - ✓ Licencing and permits
 - ✓ Environmental Regulations
 - ✓ Airspace Regulations
 - ✓ Intellectual Property Rights
 - ✚ Understanding copy right
 - ✚ Understanding licencing agreements
 - ✚ Explain Importance of legal compliance in projects
- **Allocating Resources**
 - ✓ Identifying activities
 - ✓ Selecting tools
 - ✓ Selecting equipment
 - ✓ Selecting Materials

Resources required for indicative content

Equipment	Computer, Printer, Plotter, White board, Black board
Materials	Pens, Notebook, Chalks, Markers, duster,
Tools	Books and Software,
Facilitation techniques	Trainer guide, Scenario, individual work, group discussion, presentation ● Trainer shows the learners all the processes of planning the work. The trainer provides the scenario about remote sensing project and asks learners to plan the work for that project individually and the trainer guides them and helps them in the process.
Formative assessment methods	● Written assessment ● Oral assessment ● Product assessment

Learning outcome 2: Collect Remote Sensing data		Learning hours: 30 hours	
Indicative content			
• Selecting data collection methods ✓ Identify work requirements ✓ Identify data collection methods ✚ Space borne remote sensing ✚ Airborne remote sensing ✚ Ground based remote sensing ✚ Acoustic remote sensing • Establishing Control points ✓ Identifying study area ✓ Select Reference datum ✓ Record control point data ✓ Establish control monument • Capturing Data ✓ Identifying data sources ✓ Selecting data format ✓ Downloading data			
Resources required for the indicative content			
Equipment		GNSS Receivers, PPE, Computer, Printer, Plotter, Digital Cameras, White board	
Materials		Batteries, pens and Wooden pegs, Cartridge, water, cement, sand	
Tools		Tape measure, Hoe, Software, Notebook, driver, Hammer and Paint brushes, Umbrella, aerial photos, ground control point (GCP)	
Facilitation techniques		Trainer guide, individual work, group work, presentation • The trainer provides scenario about photogrammetry data collection project, and guide the learners to establish the control point on the site. • The trainer presents to the learners about remote sensing data collection methods with some examples and guide them to download satellite images.	
Formative assessment methods		• Oral assessment • Performance assessment • Product assessment	

Learning outcome 3: Produce maps and models**Learning hours: 30 hours****Indicative content**

- **Pre-processing data**
 - ✓ Extracting imageries
 - ✚ Photographs
 - ✚ Satellite imagery
 - ✓ Radiometric correction
 - ✓ Atmospheric correction
 - ✓ Image Geo-referencing
- **Processing Data**
 - ✓ Applying image classification methods
 - ✚ Supervised classification
 - ✚ Unsupervised classification
 - ✓ Interpreting image
 - ✓ Assessing accuracy
- **Generating maps and models**
 - ✓ Digital Elevation Models
 - ✓ Maps
 - ✓ Customizing layout
 - ✓ Exporting maps and models
 - ✓ Plotting the maps and models

Resources required for the learning outcome

Equipment	GNSS Receivers, PPE, Computer, Printer, Plotter, and Digital Cameras
Materials	Batteries, pens, Wooden pegs, cartridge
Tools	Tape measure, Hoe, Software, Notebook, driver, Hammer and Paint brushes, water, Umbrella
Facilitation techniques	Trainer guide, individual work, presentation, photos and videos show ● Trainer shows pictures and videos on how to process remote sensing and photogrammetric data and asks learners to highlight their observation individually. ● Trainer gives the work of data extraction by using remote sensing and photogrammetric products ● Trainer guides learners to plot and export maps and models generated.
Formative assessment methods	● Oral assessment ● Performance assessment ● Product assessment

Integrated/Summative assessment

Integrated situation

Nyarugenge District is facing the issue of informal settlement development, where some households are living in high-risk areas. As Geospatial Technician, you are hired by Nyarugenge District to conduct terrain analysis, and produce thematic maps of households living in high-risk zones with more than 40% of gradient.

All required materials, tools and equipment are available, and the task should be completed in 10 hours.

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
1. Plan the work (30%)	1.1. Remote sensing systems are properly identified based on the work requirement	1. Element of remote sensing are identified			2
		2.Type of remote sensing sensors are identified			2
		3. type of remote sensing platforms are identified			2
		4. type of remote photogrammetric surveying are identified			2
	1.2. Site Reconnaissance is properly conducted based on the environmental conditions	1. Site is prepared			2
		2. Site is conducted			2
		4. Site reconnaissance report is elaborated			3
	1.3 Legal and regulatory requirements are properly described according to their types	1. Licensing and permits are identified			2
		2. Environmental regulations are identified			2
		3. Airspace regulations are identified			2

		4. Intellectual property right are identified			2
	1.4. Resources are properly allocated according to planned activities.	1. Activities are identified			2
		2. tools are selected			2
		3. Equipment t are selected			2
		4. Materials are selected			1
2. Collect remote sensing data (30%)	1. Data collection methods are correctly selected according to the work requirements	1. work requirements are identified			2
		2. Data collection methods are identified			3
	2. Control points are accurately established based on the scope of the work	1. Study area is identified			3
		2. Reference datum is select			4
		3. Control point data are recorded			4
		4. Control monuments are established			4
	3. Data are accurately captured based on the selected methods	1. Data source are identified			4
		2. Data format are selected			3
		3. Data are downloaded			3
3. Produce maps and models (40%)	1. Data are accurately pre-processed according to the pre-processing procedures	1. Images are extracted			2
		2. Radiometric correction are performed			3
		3. Atmospheric correction are performed			3
		3. Images are georeferenced			3

	4.2. Data are accurately processed according to the types of products to be generated	1. Image classification methods are applied			3
		2. Images are interpreted			4
		3. accuracy is assessed			4
	4.3. Maps and Models are accurately produced according to their format	1.Digital Elevation model are created			4
		2. Maps are created			3
		3. Layouts are customized			3
		4. Maps and Models are exported			4
		5. Maps and Models are Plotted			4
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References

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